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D.R.E.A.M: diabetes risk via explainable AI modeling

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Abstract
Most machine learning models for diabetes prediction rely on small, homogeneous datasets and fixed thresholds, producing binary outputs with limited clinical utility. These approaches lack generalizability, probabilistic awareness, and interpretability, which are essential for real-world healthcare adoption. We present Diabetes Risk via Explainable AI Modeling (D.R.E.A.M.), a framework for Type 2 diabetes mellitus (T2DM) risk prediction that delivers continuous, calibrated probabilities with transparent explanations. D.R.E.A.M. integrates two complementary datasets (PIMA and BRFSS 2015) after excluding gestational diabetes cases, applies clinically guided feature engineering and class balancing, and trains ensemble models (Random Forest, XGBoost, LightGBM). Decision thresholds are optimized using precisionâ€“recall curve analysis rather than default cutoffs, enabling clinically meaningful stratification. Model interpretability is achieved through SHapley Additive exPlanations (SHAP), providing both global and patient-level insights. All models achieved Area Under the Curves above 0.83 and F1-scores of 0.78, with Random Forest offering the best balance of sensitivity (recall = 0.89 at an optimized threshold of 0.389) and interpretability. SHAP confirmed the contribution of both physiological and behavioral factors, including glucose, BMI, blood pressure, cholesterol, and physical activity. Accessible via a lightweight web interface, D.R.E.A.M. provides real-time, explainable risk scores to support personalized preventive strategies. In summary, D.R.E.A.M. advances beyond conventional post-hoc explainability by integrating calibrated probabilistic predictions, PRC-based thresholding, and direct clinician-facing deployment. This combination transforms it from a research prototype into a transparent and clinically actionable decision support system.

Keywords Type 2 diabetes · Risk stratification · Explainable artificial intelligence · Threshold optimization · Clinical decision support

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1 Introduction

Type 2 diabetes mellitus (T2DM) is a major global health challenge, affecting over 400 million individuals worldwide [1]. Its asymptomatic progression and delayed diagnosis often lead to severe complications, including cardiovascular disease, neuropathy, and kidney failure [2]. As a result, early identification of individuals at risk is essential for effective prevention strategies [3].

Traditional screening tools rely on rigid scoring systems with fixed cutoffs, which fail to adapt to population diversity or provide individualized assessments. Recent advances in machine learning (ML) have shown potential in modeling complex, nonlinear relationships among clinical and behavioral variables, with studies on datasets such as the PIMA Indians Diabetes Database reporting promising performance [4, 5]. However, existing ML-based approaches remain constrained by limited datasets, reliance on arbitrary thresholds (e.g., 0.5), and insufficient interpretability, all of which hinder clinical adoption [6–8].

To address these limitations, we introduce D.R.E.A.M. (Diabetes Risk via Explainable AI Modeling), a framework designed to deliver probabilistic and interpretable predictions of T2DM risk. Unlike conventional black-box approaches, D.R.E.A.M. integrates calibrated probabilities, optimized thresholds, and explainable outputs into a lightweight, clinician-facing web interface. This combination bridges the gap between technical performance and clinical usability, supporting more personalized preventive strategies.

The remainder of this paper is structured as follows: Section 2 reviews related work on ML approaches for diabetes prediction. Section 3 describes the methodology, including dataset integration, feature engineering, threshold calibration, and model explainability. Section 4 presents the experimental results and discusses model performance and clinical implications. Finally, Section 5 concludes the paper and outlines future directions.

2 Related work

Despite the abundance of research on diabetes prediction using ML, most existing approaches suffer from critical and recurring limitations that compromise their clinical applicability. Much of the literature relies almost exclusively on the PIMA Indian Diabetes Dataset (PIDD), which, although widely used, represents a small, demographically homogeneous sample of female patients from a specific ethnic group. Consequently, models trained solely on this dataset are inherently limited in generalizability and clinical transferability to broader, more diverse populations.

Furthermore, almost all of these models, both those based on classical algorithms such as decision trees and Naïve Bayes and those based on more advanced techniques such as RF and XGBoost, emit binary predictions (diabetic or not) based on a default threshold of 0.5. This is problematic because it does not reflect the continuous risk that characterizes complex chronic diseases such as T2DM. The Sisodia and Sisodia study [5] compares different classifiers on the PIMA data, reporting accuracy metrics and Area Under the Curve (AUC) without attempting to expose, calibrate, or interpret predicted probabilities. This binary framing is reductive and potentially harmful in clinical practice, where risk stratification and early intervention are essential.

More sophisticated works, such as those involving evolutionary optimization or metaheuristic enhancement of artificial neural networks [9], focus on improving predictive performance metrics but completely disregard the issue of transparency. These models act as black boxes: even if they achieve high accuracy, their internal logic remains opaque, offering no insight into why a specific prediction was made, or what an unacceptable trade-off is in high-stakes domains like healthcare. Explainability methods, such as SHAP, have begun to appear in studies, including the one by Chang et al. [10], which integrates SHAP into a diabetes prediction model. However, such efforts remain superficial and purely descriptive. The explanations do not influence thresholding, calibration, or clinical decision logic. The model outputs remain binary and disconnected from any actionable, continuous risk representation. SHAP is treated as a post-hoc add-on, not as an integral part of a trustworthy pipeline.

Recent approaches have begun to explore the use of Large Language Models and conversational agents for diabetes diagnosis, such as the DiabeTalk chatbot developed using WizardLM [11]. While promising in terms of interactivity and user engagement, these models currently underperform compared to traditional supervised architectures like WE-LSTM networks, particularly in accuracy and specificity, highlighting the need for clinically fine-tuned foundations.

Another glaring omission across these studies is the lack of *probability calibration*. Guo et al. [12] convincingly demonstrate that even high-performing models often produce poorly calibrated probabilities, that is, predicted scores that do not align with real-world confidence levels. In the context of diabetes prediction, this undermines the clinician's ability to interpret risk levels and make informed decisions. Most ML-based studies ignore this calibration gap entirely, relying on unvalidated or arbitrarily chosen thresholds.

Equally concerning is the near-total absence of appropriate evaluation for *imbalanced datasets*. In the presence of class imbalance common in real-world screening tasks, the Receiver Operating Characteristic (ROC) curve can be misleading. Saito and Rehmsmeier [13] demonstrate that PRCs provide a more accurate evaluation of model performance under such conditions. However, PRC-based threshold optimization is practically non-existent in the literature, despite its clear clinical relevance.

Lastly, no prior work integrates these components into a system usable in practice. Most models remain as prototypes, with no web-based or clinician-facing interface. There is a critical gap between technical potential and real-world utility (Table 1).

D.R.E.A.M. addresses all these shortcomings in a unified, clinically oriented framework. It combines two heterogeneous datasets PIMA and Behavioral Risk Factor Surveillance System (BRFSS), ensuring broader population coverage and consistency in defining T2DM by excluding gestational cases. It applies clinically inspired feature interactions, balances class distributions using SMOTE, and trains robust ensemble models with calibrated probabilities. Most importantly, it abandons the rigid threshold of 0.5, instead deriving clinically meaningful cutoffs based on PRC optimization for decision-making. Model results are fully interpretable through SHAP as an integral component of the analysis. These results are distributed through a web-based interface, which enables real-time risk prediction that is explainable and accessible to clinicians. In this way, D.R.E.A.M. transforms ML for diabetes from experimental to actionable, offering personalized, calibrated, and transparent decision support.

Table 1 Summary of representative ML-based studies for diabetes prediction, highlighting key characteristics and limitations

Author (Year)	Dataset(s)	Method / Model	Limitations
Sisodia and Sisodia [5]	PIMA	Decision trees, SVM, Naïve Bayes	Binary outputs only, no calibration, limited generalizability (PIMA only).
Jahani et al. [9]	PIMA	ANN with evolutionary optimization	Focus on accuracy only, lacks transparency and interpretability.
Chang et al. [10]	PIMA	ML + SHAP explanations	SHAP used superficially as post-hoc, thresholds remain arbitrary (0.5).
Guo et al. [12]	Various	Calibration study across ML models	Highlights poor calibration, but not applied to diabetes-specific pipeline.
Saito and Rehmsmeier [13]	Simulation	PRC analysis	Demonstrates PRC value, but not integrated in diabetes prediction.
DiabeTalk (2024) [11]	Custom	LLM-based chatbot	Promising interactivity but low accuracy and specificity.

3 Methodology

The D.R.E.A.M. framework was designed to deliver a clinically robust, probabilistically interpretable, and generalizable solution for predicting T2DM risk. The methodology comprises six main stages: dataset integration and preprocessing, clinically guided feature engineering, class balancing, model training with threshold calibration, probability calibration, and model explainability via SHAP (Fig. 1).



Fig. 1 Architecture of the entire D.R.E.A.M. execution pipeline

3.1 Dataset integration and harmonization

We integrated two complementary datasets: the PIMA Indian Diabetes Dataset (PIMA) and the BRFSS 2015.

PIMA consists of clinical measurements collected from female patients of Pima Indian heritage and is widely used in diabetes prediction research. The BRFSS, by contrast, is a large-scale, population-level health survey that is representative of the general U.S. adult population. To ensure consistency in modeling T2DM, we excluded cases potentially indicative of gestational diabetes by retaining only non-pregnant individuals (`Pregnancies = 0`) from the PIMA dataset. For BRFSS, we leveraged its explicitly labeled binary outcome variable (`Diabetes_binary`), which already excludes gestational and prediabetic cases.

To harmonize the two datasets, we identified a standard set of clinical and behavioral features and aligned their naming conventions. Variables missing in either source were initially set as `NaN` to preserve structural compatibility. Before imputation, we filtered out rows with excessive missingness (thresh = 5), ensuring sufficient information per record. The remaining missing values were imputed using the median of each variable, computed across the entire combined dataset. This harmonization strategy allowed us to retain valuable diversity across populations while ensuring a coherent and analyzable feature space for downstream modeling. Overall, the PIMA dataset comprises 768 samples, of which 268 (35%) are positive for T2DM and 500 (65%) are negative. The BRFSS 2015 survey includes 441,456 respondents, with an overall diabetes prevalence of approximately 11%. After harmonization, the combined dataset reflects both a small, clinically detailed cohort (PIMA) and a large, heterogeneous population (BRFSS), capturing complementary perspectives while also highlighting the class imbalance inherent in real-world screening tasks.

3.2 Clinically inspired feature engineering

We augmented the dataset with interaction-based features derived from domain-specific clinical reasoning. These engineered variables aim to capture latent, nonlinear relationships among physiological markers known to modulate metabolic risk [14, 15]. Specifically, we introduced: `BMI_Age` (Body Mass Index multiplied by Age), `Glucose_Age`, `BMI_Glucose`, `BP_Age` (systolic/diastolic proxy times Age), and `BP_Chol` (interaction between BP and cholesterol levels).

The interaction features engineered in this study were not arbitrarily selected but rather grounded in established clinical evidence regarding the pathophysiology of T2DM.

For example, the Glucose Ã— Age variable captures the increasing difficulty in glycemic control associated with aging, due to progressive β-cell dysfunction and insulin resistance observed in elderly individuals [16, 17]. Similarly, BMI $\times$ Age accounts for the compounding effect of long-term obesity, as prolonged adiposity increases chronic inflammation and metabolic burden, accelerating the onset of insulin resistance with age [18].

The interaction between BP and Cholesterol (BP Ã— Cholesterol) reflects their synergistic role in cardiometabolic risk. Hypertension and dyslipidemia are two major components of the metabolic syndrome and are strongly predictive of both macrovascular and microvascular diabetic complications [19, 20]. Including these interactions allows the model to better capture latent, non-linear risk patterns well-documented in clinical endocrinology.

Similarly, `BP_Chol` approximates cardiovascular burden, which is tightly linked to insulin resistance and metabolic syndrome.

To reduce potential redundancy and multicollinearity, we computed pairwise correlations and feature importances from an initial RF baseline to ensure that each engineered feature contributed independently to the model. Although no formal dimensionality reduction technique was employed, this post-hoc analysis guided feature retention and helped avoid overfitting. To quantify the impact of engineered features, we conducted an ablation study (Table 2) comparing model performance with and without interaction terms (BMI $\times$ Age, Glucose $\times$ Age, BP $\times$ Chol). Removing these features decreased F1-score (from 0.78 to 0.72) and recall (from 0.89 to 0.81), indicating their substantial contribution to model sensitivity and overall predictive power.

3.3 Class imbalance handling

Given the natural bias in class distribution, confirmed by a preliminary study of the datasets, we applied SMOTE to mitigate class imbalance before training. SMOTE generates new synthetic samples by interpolating between existing instances of the minority class, thus avoiding the overfitting typically introduced by simple duplication.

Balancing the dataset in this way is particularly important in the context of chronic disease prediction, where false negatives can have serious clinical consequences. Without this correction, models tend to favor the majority class, resulting in high overall accuracy but poor sensitivity and recall for diabetic cases. By applying SMOTE before the split between training and testing, we ensured that both the training and evaluation phases reflected a balanced distribution of classes, allowing classifiers to better generalize and detect patterns of minority classes. This step was crucial for enabling fair threshold optimization and reliable performance assessment across all models. In terms of class distribution, before applying SMOTE the positive (diabetic) class accounted for approximately 11% of the combined dataset, with the negative class representing about 89%. After SMOTE, the dataset was rebalanced to achieve an even 50/50 distribution between classes.

3.4 Model training, validation, and threshold optimization

We evaluated three ensemble classifiers: RF, XGBoost, and LightGBM.These models were selected based on their proven effectiveness in biomedical prediction tasks. Random Forest is widely used for its robustness to noise and ease of interpretability, which are key characteristics for clinical adoption [21, 22]. XGBoost has consistently demonstrated state-of-the-art performance on structured tabular healthcare data, as reported in previous studies [23, 24]. LightGBM was included for its computational efficiency and scalability on large datasets, making it suitable for real-world implementation [25, 26]. Each model was embedded in a `Pipeline` that included `StandardScaler` normalization to ensure feature comparability. To improve robustness and generalizability, we employed 5-fold cross-validation

Table 2 Ablation study on engineered interaction features. Metrics are averaged across 5-fold cross-validation

Configuration	F1-score	Recall
Full model (with engineered features)	0.78	0.89
Without engineered features	0.72	0.81

on the training set during model selection and hyperparameter tuning, and we reported mean performance metrics across folds.

Rather than using the conventional 0.5 decision threshold, we optimized thresholds by analyzing the PRC on the validation folds to maximize the F1-score. This is particularly effective in imbalanced settings where ROC curves may be misleading [13]. For instance, the optimal threshold for RF was found to be 0.389, allowing earlier identification of at-risk patients who would not be flagged under standard cutoffs. We employed default hyperparameters across all classifiers. This was an intentional design choice to ensure a fair and reproducible comparison between algorithms, avoiding bias introduced by extensive hyperparameter tuning. By maintaining default settings, performance differences can be more directly attributed to the model architectures themselves rather than to parameter optimization. Future work will systematically explore hyperparameter search strategies (grid or random search) to further refine model performance.

3.5 Probability calibration

While ensemble models such as RF and XGBoost natively output class probabilities, these raw scores are not always well-calibrated, particularly when used for clinical interpretation. We applied post-training probability calibration using isotonic regression, fitted on the validation folds to address this. This step ensures that predicted probabilities more accurately approximate actual outcome likelihoods, a crucial requirement for any risk stratification tool intended to inform clinical decision-making. The calibrated outputs were then used to identify optimal classification thresholds and to generate interpretable risk scores.

3.6 Model explainability and output accessibility

To enhance transparency and promote clinical trust, we applied SHAP to the best-performing model, RF, to extract global and patient-level feature attributions. SHAP values were computed on a held-out test subset and visualized through summary plots and waterfall charts. These insights were then integrated directly into the user-facing interface, enabling clinicians to view a probabilistic risk score for each patient and understand the contribution of individual clinical features driving that prediction.

The complete D.R.E.A.M. system is accessible via a lightweight web-based interface that supports real-time predictions. Clinicians can input patient-specific data and instantly receive a calibrated risk score and visual SHAP-based explanations. This seamless end-to-end integration from raw input to interpretable output represents AI's practical and trustworthy deployment in real-world diabetes screening scenarios.

3.7 Web interface and deployment

To bridge the gap between technical development and clinical application, we developed a lightweight, web-based interface for `D.R.E.A.M.` (Fig. 2). The interface enables healthcare professionals to input individual patient data including clinical parameters such as BMI, Glucose, BP, and behavioral variables such as smoking and physical activity status via an intuitive form equipped with sliders and radio buttons.

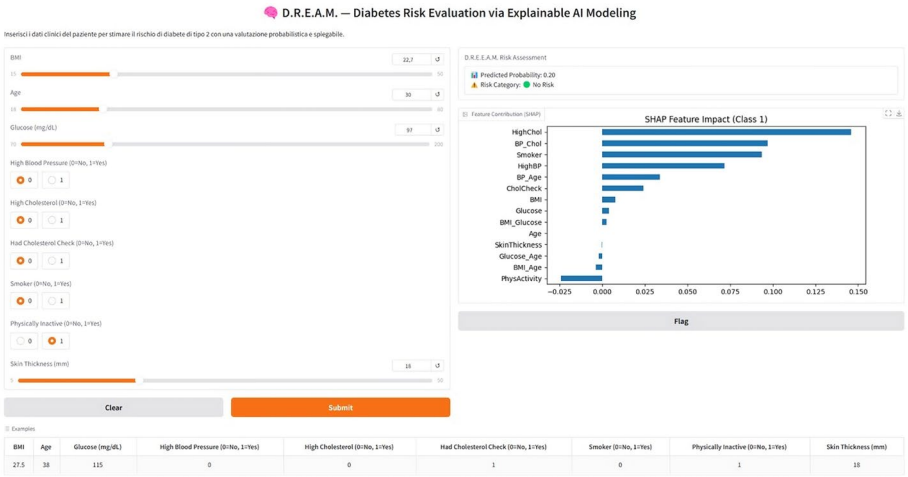


Fig. 2 D.R.E.A.M. system interface. The web application enables diabetes risk estimation and provides a visual explanation through SHAP, offering clinicians actionable and interpretable feedback

After data entry, the system uses the calibrated Random Forest model to compute the probability of T2DM and assigns the output to one of four clinically meaningful risk categories [27]:

- No Risk: $p < 0.25$
- Low Risk: $0.25 \leq p < 0.389$
- Moderate Risk: $0.389 \leq p < 0.7$
- High Risk: $p \geq 0.7$

This granular stratification supports personalized preventive strategies, moving beyond rigid binary classification schemes. Additionally, the interface generates real-time SHAP-based visualizations to explain the contribution of each feature to the predicted risk. This empowers clinicians to understand which factors, such as elevated glucose, high BMI, or lack of physical activity, most strongly influence the modelâ€™s decision.

The system is fully web-accessible, mobile-friendly, and requires no installation or credentials, making it a practical solution for real-world deployment in clinical and telemedicine contexts.

4 Results and conclusion

The three ensemble classifiers RF, XGBoost, and LightGBM achieved strong predictive performance on the harmonized PIMA+BRFSS dataset, with AUC values above 0.83 and F1-scores around 0.78. To test generalizability, we conducted external validation on a subset of the NHANES dataset (2017â€“2020), where the calibrated RF model maintained robust performance (AUC = 0.826, F1 = 0.75) [28].

Although all models performed comparably in overall metrics, Random Forest emerged as the most clinically appropriate choice for integration into the D.R.E.A.M. framework. RF

achieved a recall of 0.89 and precision of 0.70 at an optimized threshold of 0.389 (derived via precision-recall curve analysis), enabling earlier identification of high-risk individuals compared to the conventional 0.5 cutoff. This high sensitivity aligns with clinical priorities where minimizing false negatives is essential.

Moreover, RF outperformed gradient boosting methods in interpretability. Thanks to its tree-based structure and integrated SHAP (Fig. 3) analysis, RF provides transparent and clinically intuitive explanations at both the global and patient level—critical for fostering trust in decision support systems (Table 3).

The reported AUC and F1 values represent the mean performance across 5-fold cross-validation. While full bootstrap-based confidence intervals could not be computed in this revision, the use of cross-validation already provides an estimate of internal variability and robustness. We acknowledge this as a limitation and plan to include explicit confidence intervals from external validation cohorts in future work.

In conclusion, D.R.E.A.M. presents a clinically explainable, web-based decision support system for early T2DM risk assessment. It combines calibrated probabilistic predictions, interpretable outputs via SHAP, and a stratified risk scoring scheme to guide personalized interventions. The system bridges the gap between machine learning performance and real-world clinical applicability. We also acknowledge that practical challenges remain for large-

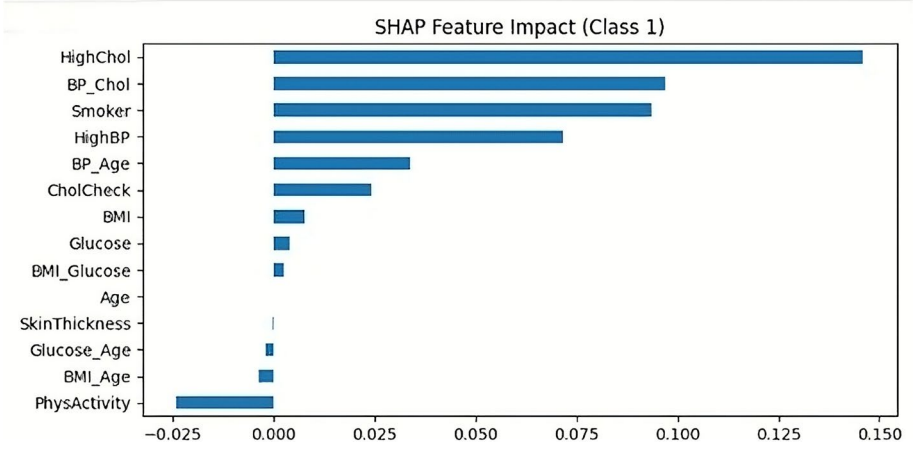


Fig. 3 SHAP summary plot illustrating the relative contribution of features to Type 2 diabetes risk prediction. Higher values indicate a stronger impact on model output, confirming the importance of physiological (glucose, BMI, blood pressure, cholesterol) and behavioral (smoking, physical activity) factors

Table 3 Performance metrics for each classifier. Metrics are reported on the test set, using model-specific thresholds derived from PRC analysis. External validation results on NHANES are included for RF

Metric	RF	XGBoost	LightGBM	RF (NHANES)
Accuracy	0.75	0.74	0.74	0.74
F1-score	0.78	0.78	0.78	0.75
AUC	0.8417	0.8391	0.8377	0.826
Average precision	0.8282	0.8309	0.8298	0.8024
Precision	0.70	0.68	0.69	0.71
Recall	0.89	0.90	0.89	0.80
Optimal threshold	0.389	0.366	0.381	0.389*

*Threshold from internal PRC used on external NHANES set

scale deployment, including data privacy, integration with existing electronic health record systems, and the need for clinician training to ensure proper adoption and trust. Future work will focus on prospective validation, EHR integration, and application to other chronic disease contexts.

5 Discussion and limitations

An important clinical implication of our work concerns the optimized decision threshold. While most models rely on the conventional cutoff of 0.5, our precisionâ€“recall curve analysis identified 0.389 as the optimal value for Random Forest. This lower threshold increases sensitivity and allows earlier identification of at-risk individuals, which is particularly valuable in preventive contexts. We acknowledge that this comes at the cost of more false positives, but in chronic diseases such as T2DM, early detection outweighs additional follow-up. Moreover, SHAP analysis confirmed that features such as glucose and BMI were the main drivers of high-risk predictions, while protective factors like physical activity reduced risk estimates, further supporting the clinical interpretability of the model outputs.

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